SENSOR

NASPIER V2
IMU BOARD

File: SENSOR_IMU.kicad_sch

HIROSE 34P
CONNECTOR

FLEX_CABLE

File: FLEX_CABLE_sch.kicad_sch

CARRIER

PAB STANDARD
CARRIER BOARD

File: CARRIER.kicad_sch

CORE

File: CORE.kicad_sch

| | | | |
|---|---|---|---|
| Schematic | **Naspier Core** | Create at | 2026-09-20 |
| | | Update at | 2026-10-15 |
| Board | NASPIER_CORE_REV0_VER0 | Page | NASPIER CORE: SYSTEM (CORE / SENSOR / CARRIER) |
| Drawn | Ahmet Vapurcu | **Naspier Core** | |
| Reviewed | | | |
| | | Version | Size | Page 1 Total 17 |
| | | REV0 VER0 | A3 | VAP-SAN |

HIROSE 34P
FFC CONNECTOR

CORE_H743

STM32H743IIK6

ARM® Cortex®-M7 STM32H7 Microcontroller IC
32-Bit 480MHz 2MB (2M x 8) FLASH
201 UFBGA (10x10)

File: CORE_H743.kicad_sch

CORE_POWER

IMU BUSSES 3.3V
REGULATORS

FC & AUX 3.3V
REGULATORS

File: CORE_POWER.kicad_sch

CORE_ONBOARD

AT24C64D
EEPROM-I2C

FRAM
SPI

ICM-45686
IMU-SPI

MS5611
BARO-I2C

File: CORE_ONBOARD.kicad_sch

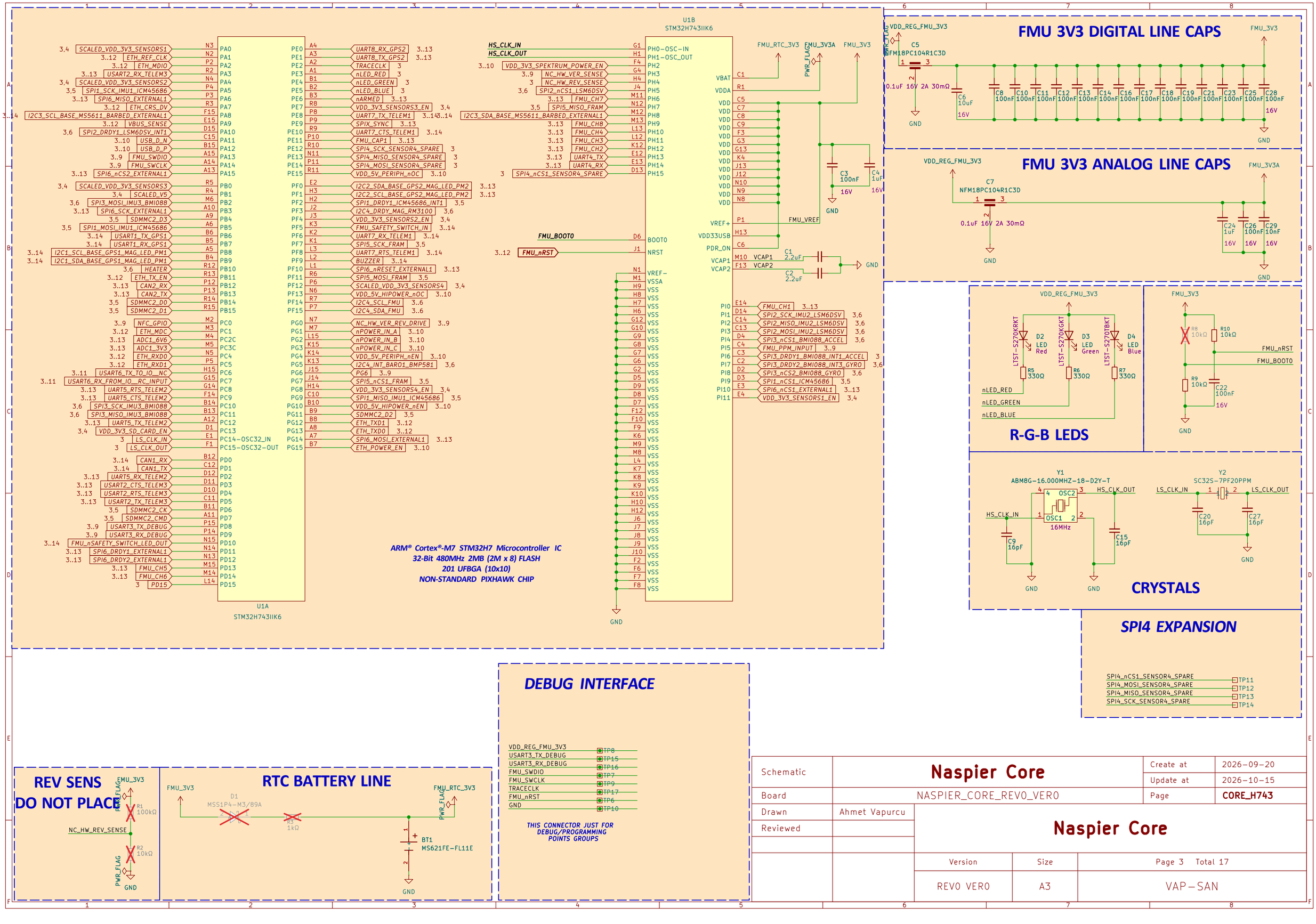
CORE_CONNECTORS

HIROSE 100P
CONNECTOR

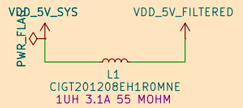
HIROSE 50P
CONNECTOR

File: CORE_CONNECTORS.kicad_sch

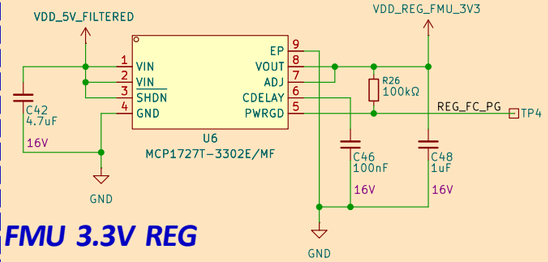
| | | | |
|---|---|---|---|
| Schematic | Naspier Core | Create at | 2026-09-20 |
| | | Update at | 2026-10-15 |
| Board | NASPIER_CORE_REV0_VER0 | Page | CORE BOARD |
| Drawn | Ahmet Vapurcu | Naspier Core | |
| Reviewed | | | |
| | | Version | Size | Page 2 Total 17 |
| | | REV0 VER0 | A4 | VAP-SAN |


U1A
STM32H743IIK6
U1B
STM32H743IIK6
ARM® Cortex®-M7 STM32H7 Microcontroller IC
32-Bit 480MHz 2MB (2M x 8) FLASH
201 UFBGA (10x10)
NON-STANDARD PIXHAWK CHIP
FMU 3V3 DIGITAL LINE CAPS
FMU 3V3 ANALOG LINE CAPS
R-G-B LEDS
CRYSTALS
SPI4 EXPANSION
REV SENS
DO NOT PLACE
RTC BATTERY LINE
DEBUG INTERFACE
THIS CONNECTOR JUST FOR
DEBUG/PROGRAMMING
POINTS GROUPS
Naspier Core
NASPIER_CORE_REV0_VER0
Ahmet Vapurcu
REV0 VER0
A3
VAP-SAN
2026-09-20
2026-10-15
CORE_H743
Page 3 Total 17

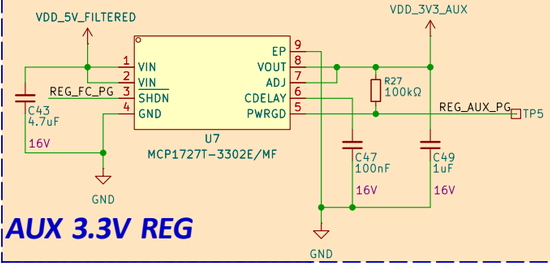
## 5V LINE FILTER



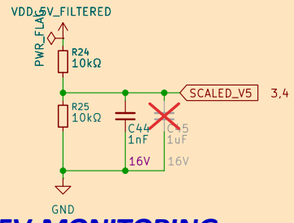

## FMU 3.3V REG



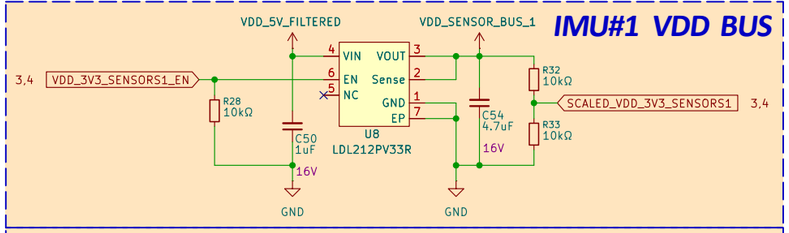

## AUX 3.3V REG



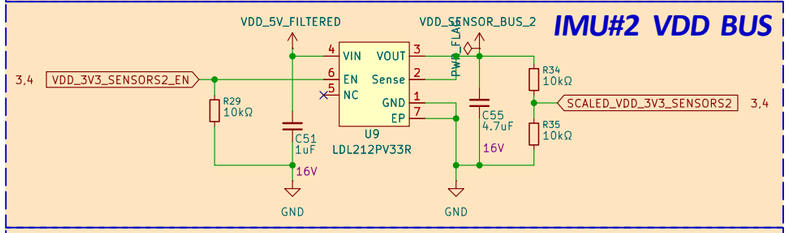

## 5V MONITORING



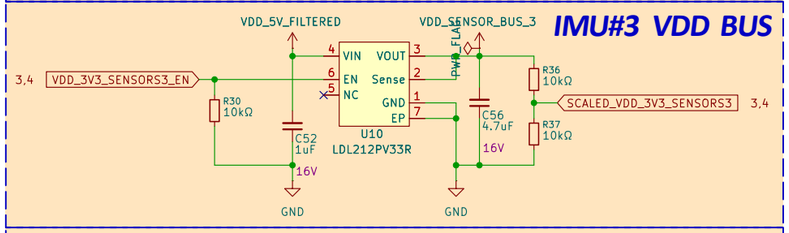

## IMU#1 VDD BUS



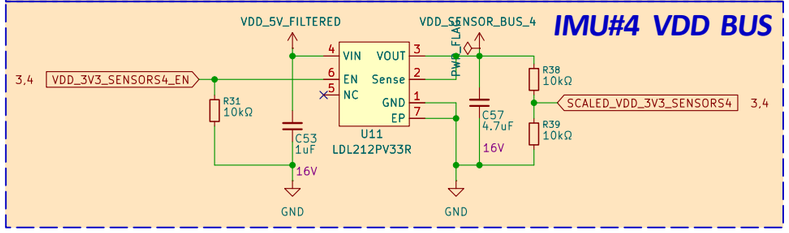

## IMU#2 VDD BUS



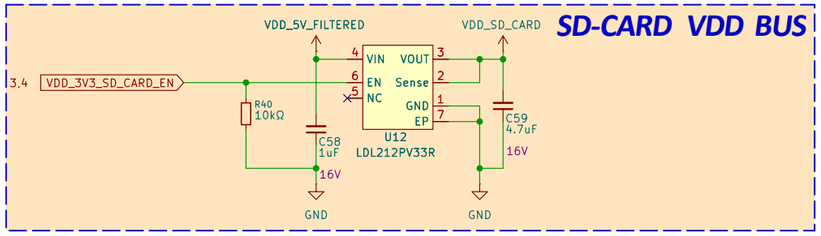

## IMU#3 VDD BUS




## IMU#4 VDD BUS

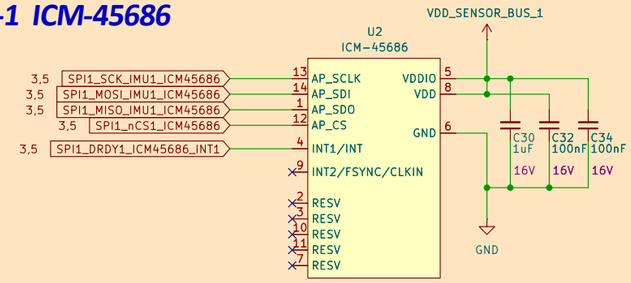



## SD-CARD VDD BUS

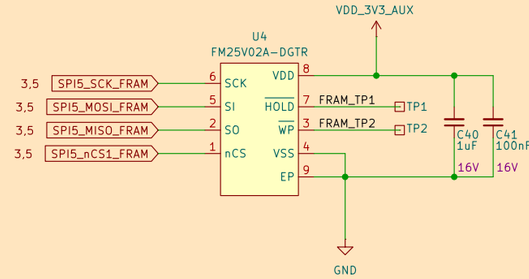



| | | | |
|---|---|---|---|
| Schematic | **Naspier Core** | Create at | 2026-09-20 |
| | | Update at | 2026-10-15 |
| Board | NASPIER_CORE_REV0_VER0 | Page | **CORE_POWER** |
| Drawn | Ahmet Vapurcu | **Naspier Core** | |
| Reviewed | | | |
| | | Version | Size / Page 4 Total 17 |
| | | REV0 VER0 | A3 / VAP-SAN |

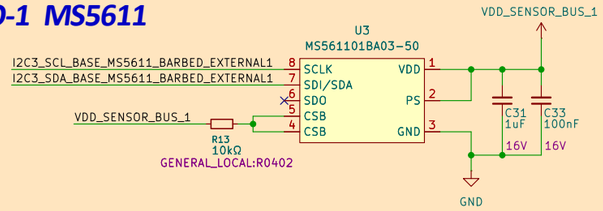
## IMU-1 ICM-45686



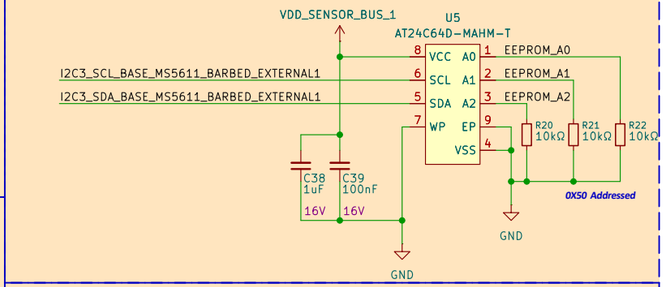

## FRAM



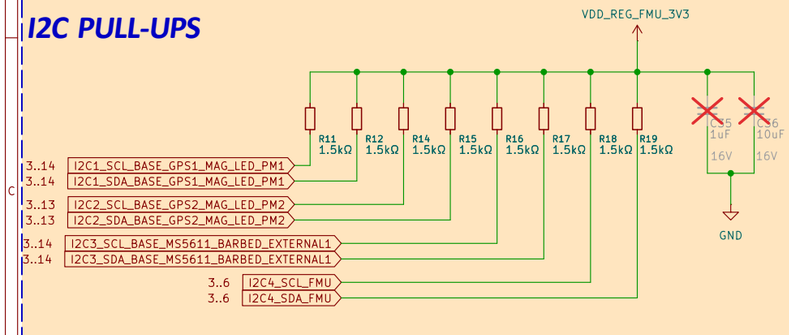

## BARO-1 MS5611



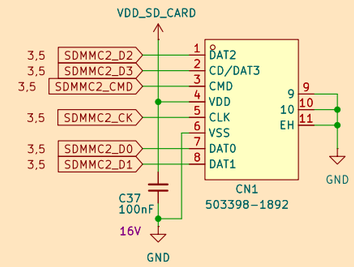

## ONBOARD EEPROM




## I2C PULL-UPS




## SD-CARD

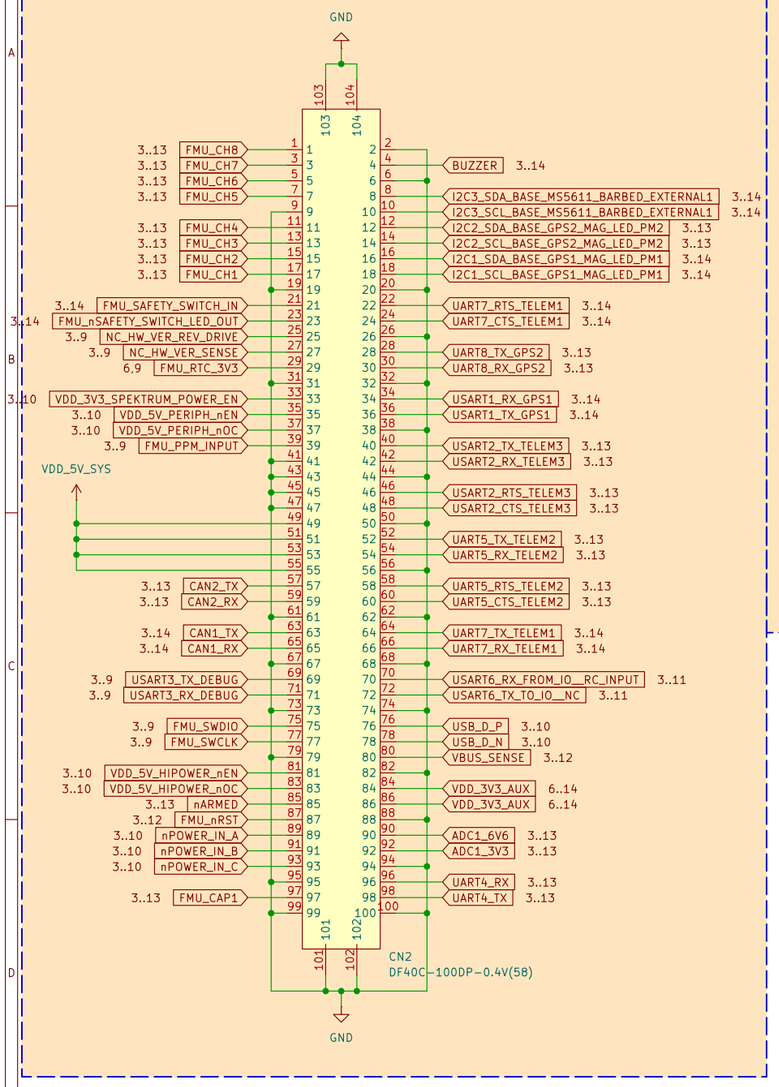



**Push-In**
**Push-Out**

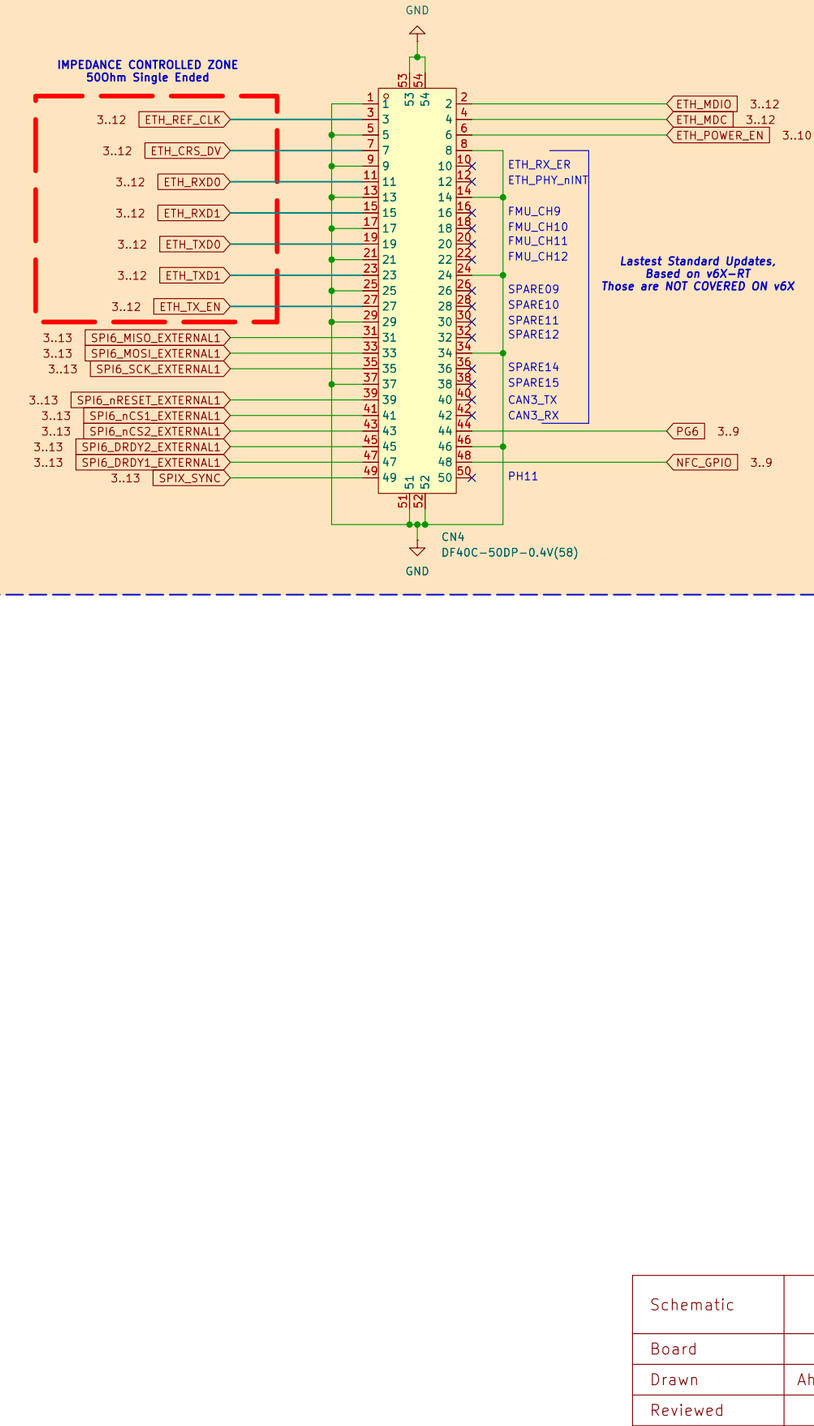
| | | | |
|---|---|---|---|
| Schematic | **Naspier Core** | Create at | 2026-09-20 |
| | | Update at | 2026-10-15 |
| Board | NASPIER_CORE_REV0_VER0 | Page | **CORE_ELEMENTS** |
| Drawn | Ahmet Vapurcu | **Naspier Core** | |
| Reviewed | | | |

| Version | Size | Page 5 Total 17 |
|---|---|---|
| REV0 VER0 | A3 | VAP-SAN |

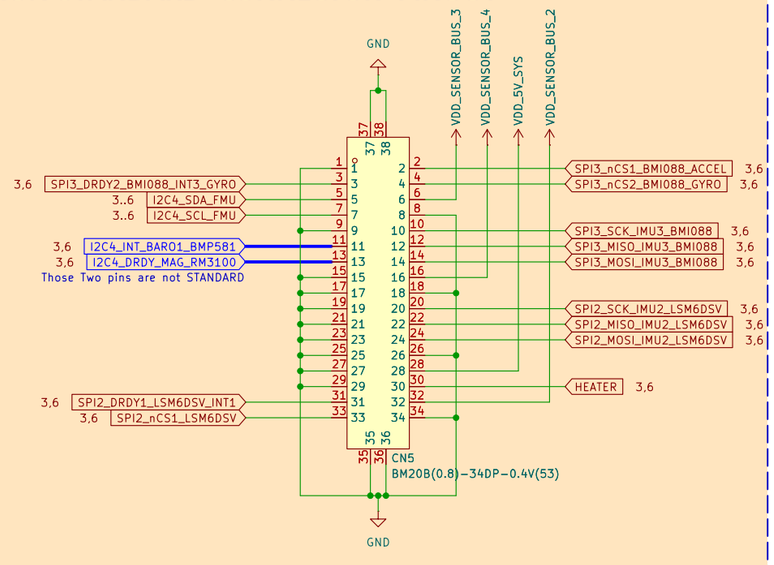
## PAB X1 100P HIROSE




## PAB X2 50P HIROSE

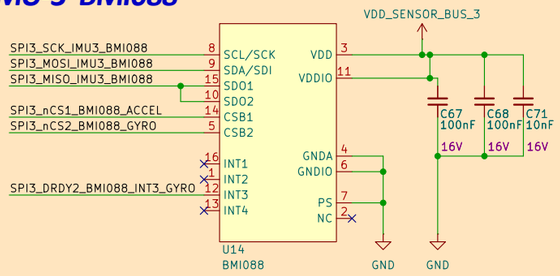



## PAB X3 34P SENSOR FFC CONNECTOR
## NON-STANDARD CONNECTOR PINOUT

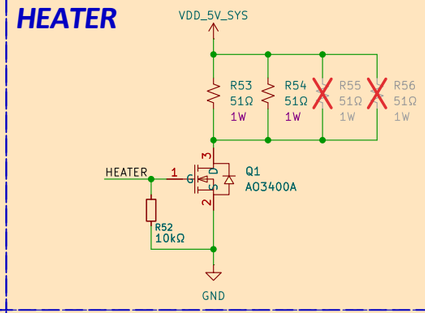



| | | | |
|---|---|---|---|
| Schematic | **Naspier Core** | Create at | 2026-09-20 |
| | | Update at | 2026-10-15 |
| Board | NASPIER_CORE_REV0_VER0 | Page | **CORE_B2B** |
| Drawn | Ahmet Vapurcu | **Naspier Core** | |
| Reviewed | | | |
| | | Version / Size / Page 6 Total 17 | |
| | | REV0 VER0 / A3 / VAP-SAN | |

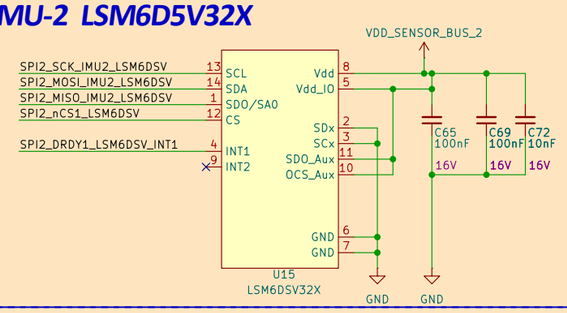
## IMU-3 BMI088



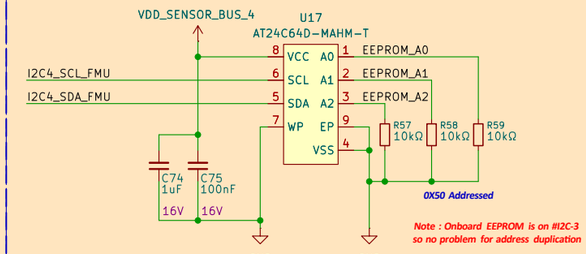

## HEATER



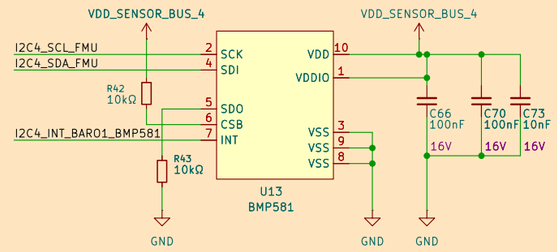

## IMU-2 LSM6D5V32X



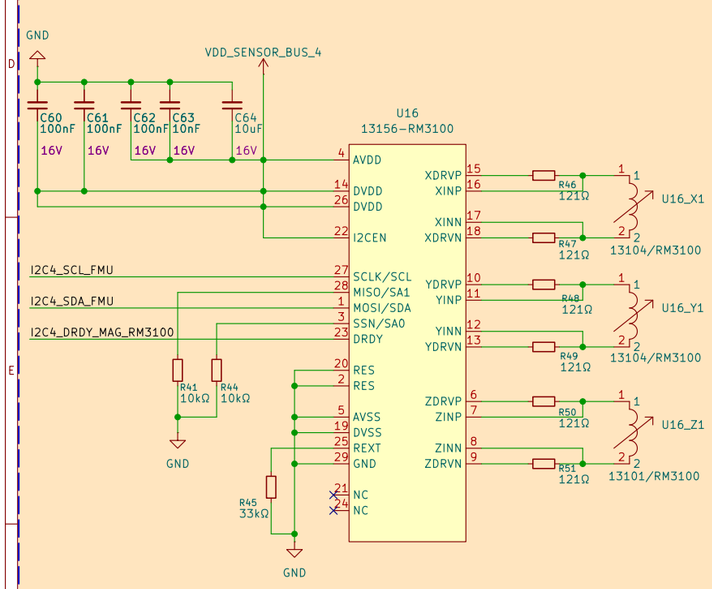

## EEPROM



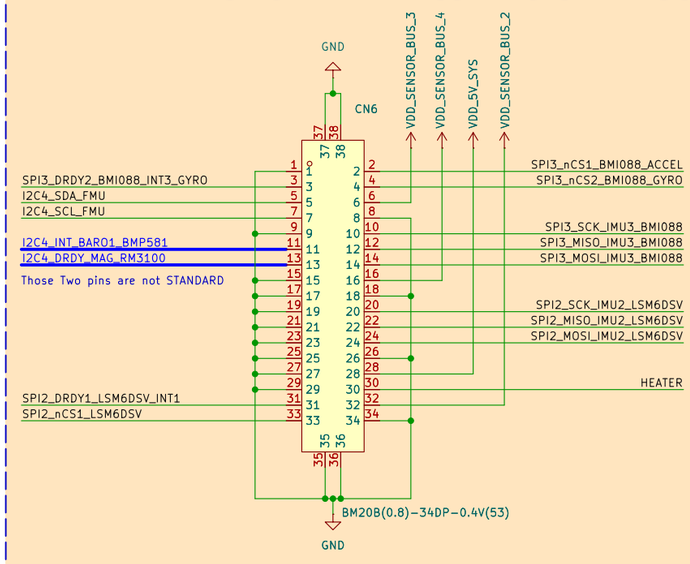

## BARO-2 BMP581




## MAG-RM3100




## PAB X3 34P SENSOR FFC CONNECTOR
## NON-STANDARD CONNECTOR PINOUT




**FOR BETTER FFC LAYOUT**
**PINOUT HERE WILL CHANGE**

| | | | |
|---|---|---|---|
| Schematic | **Naspier Core** | Create at | 2026-09-20 |
| Board | NASPIER_CORE_REV0_VER0 | Update at | 2026-10-15 |
| Drawn | Ahmet Vapurcu | Page | **SENSOR_BOARD** |
| Reviewed | | **Naspier Core** | |
| | | Version: REV0 VER0 | Size: A3 |
| | | Page 7 Total 17 | VAP-SAN |

CARRIER_B2B

File: CARRIER_B2B.kicad_sch

CARRIER_CONN_1

File: CARRIER_CONN_1.kicad_sch

CARRIER_IO_MCU

File: CARRIER_IO_MCU.kicad_sch

CARRIER_POWER

File: CARRIER_POWER.kicad_sch

CARRIER_CONN_2

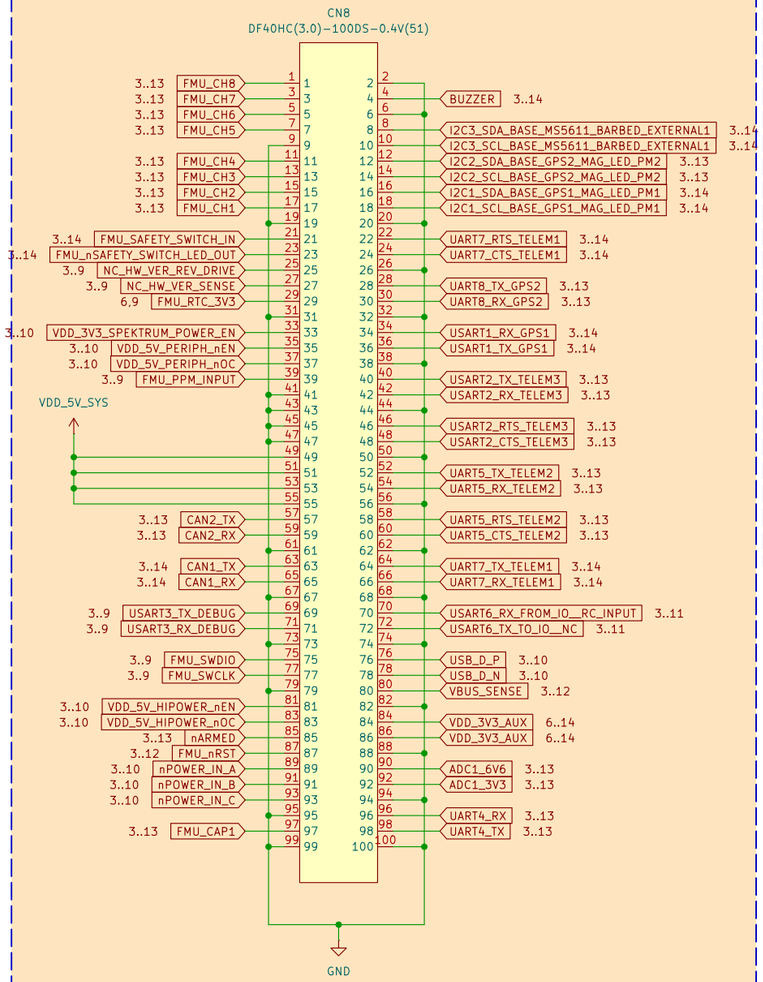
File: CARRIER_CONN_2.kicad_sch

CARRIER_ELEMENTS

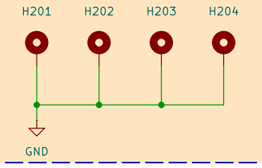
File: CARRIER_ELEMENTS.kicad_sch

| | | | |
|---|---|---|---|
| Schematic | **Naspier Core** | Create at | 2026-09-20 |
| | | Update at | 2026-10-15 |
| Board | NASPIER_CORE_REV0_VER0 | Page | **CARRIER BOARD** |
| Drawn | Ahmet Vapurcu | **Naspier Core** | |
| Reviewed | | | |
| | | Version | Size / Page 8 Total 17 |
| | | REV0 VER0 | A4 / VAP-SAN |

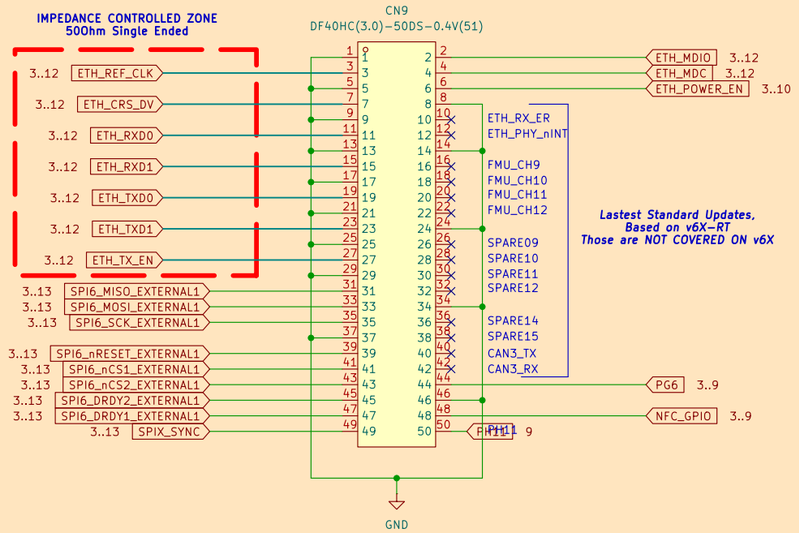
## PAB X1 100P HIROSE




## MountingHole M2 x4
## CORE SCREWED STANDOFFS

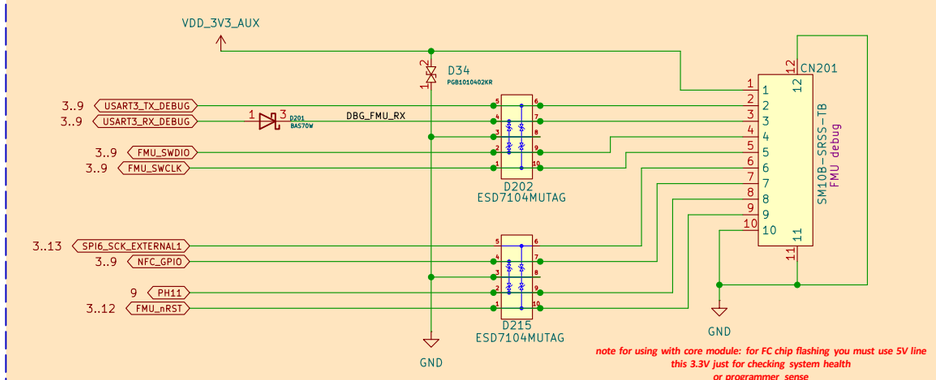



## PAB X2 50P HIROSE




Lastest Standard Updates,
Based on v6X-RT
Those are NOT COVERED ON v6X

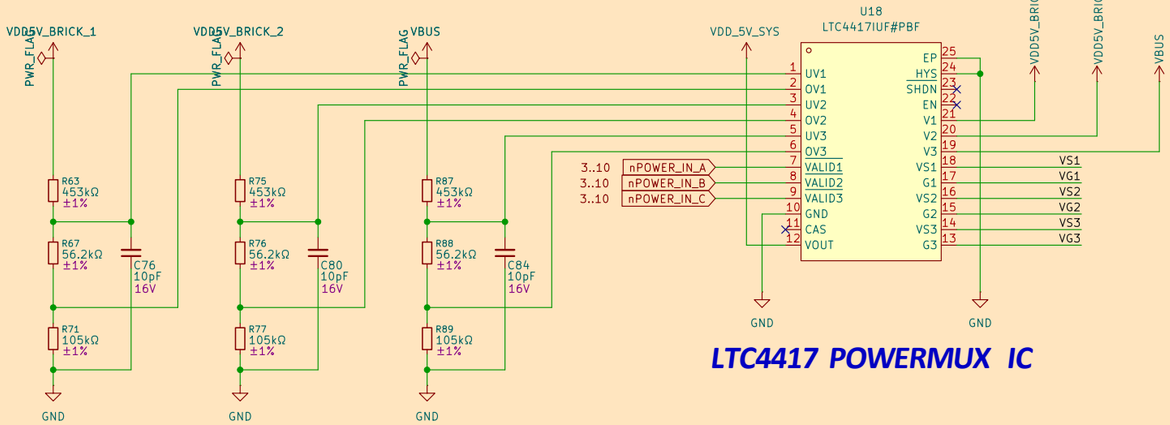
## FMU DEBUG



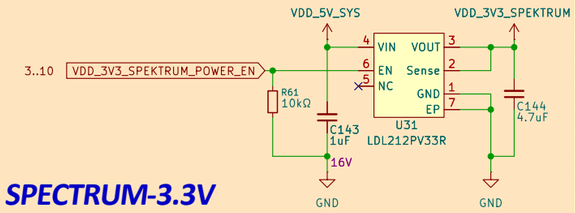

note for using with core module: for FC chip flashing you must use 5V line
this 3.3V just for checking system health
or programmer sense
Other Standard module's fc chip can be powered by this bus

| | | | |
|---|---|---|---|
| Schematic | **Naspier Core** | Create at | 2026-09-20 |
| | | Update at | 2026-10-15 |
| Board | NASPIER_CORE_REV0_VERO | Page | **CARRIER_B2B** |
| Drawn | Ahmet Vapurcu | **Naspier Core** | |
| Reviewed | | | |
| | | Version / Size / Page 9 Total 17 | |
| | | REV0 VERO / A3 / VAP-SAN | |

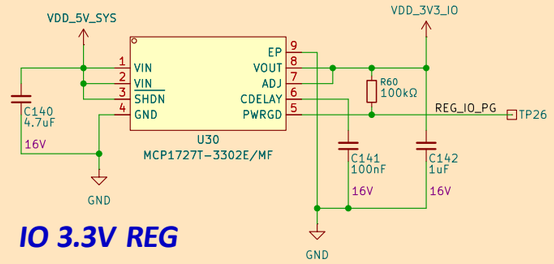
## LTC4417 POWERMUX IC



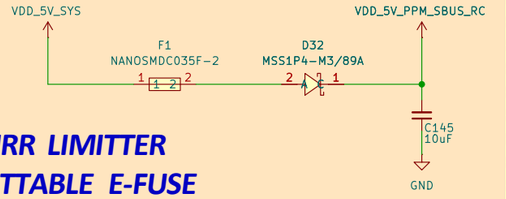

## SPECTRUM-3.3V



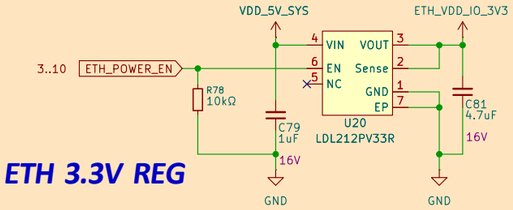

## IO 3.3V REG



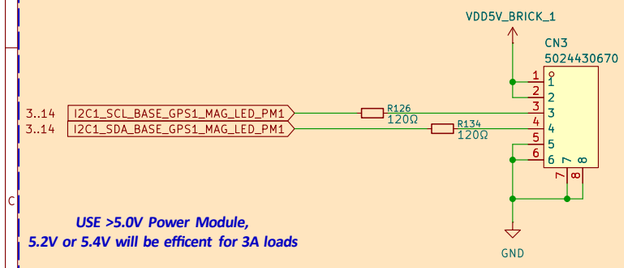

## CURR LIMITTER RESETTABLE E-FUSE




## ETH 3.3V REG

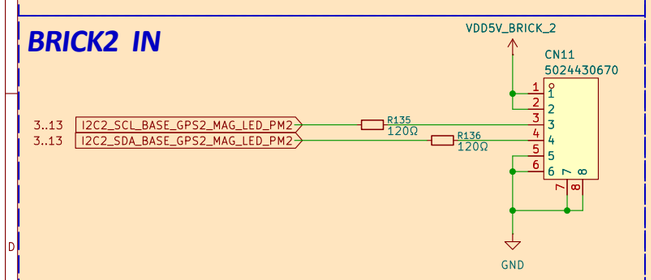



## BRICK1 IN

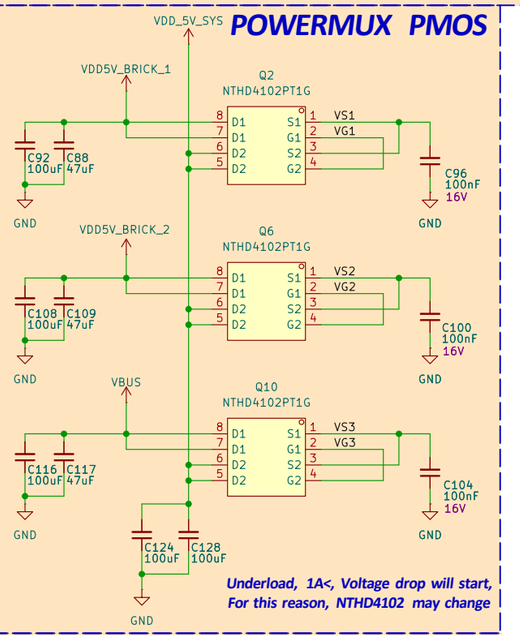



USE >5.0V Power Module,
5.2V or 5.4V will be efficent for 3A loads

## BRICK2 IN

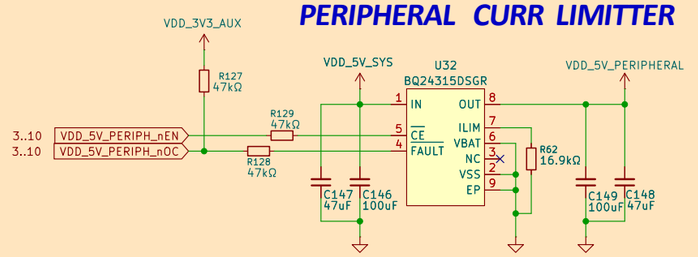



## POWERMUX PMOS

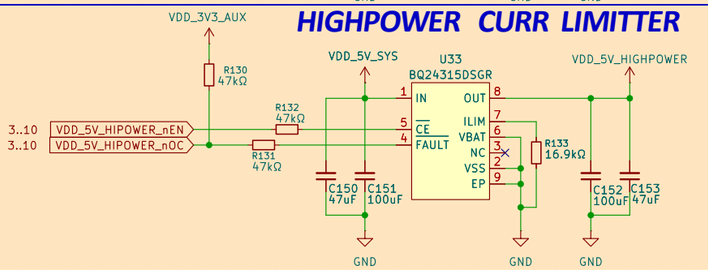



Underload, 1A<, Voltage drop will start,
For this reason, NTHD4102 may change

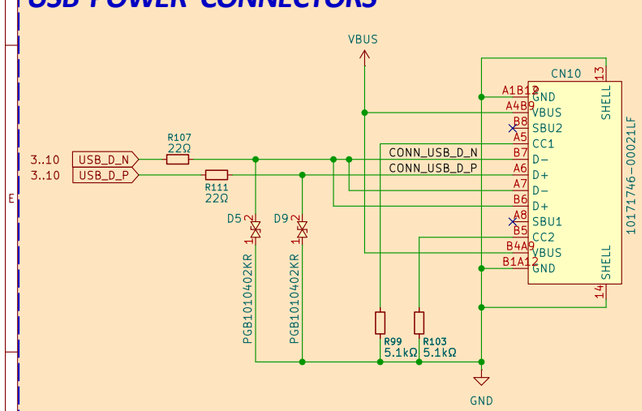
## PERIPHERAL CURR LIMITTER

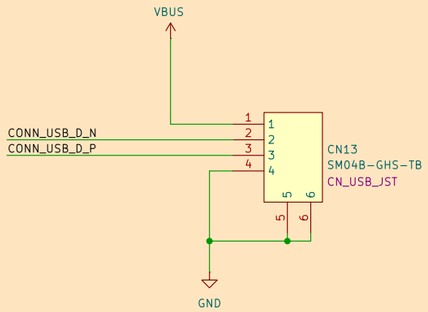



## HIGHPOWER CURR LIMITTER



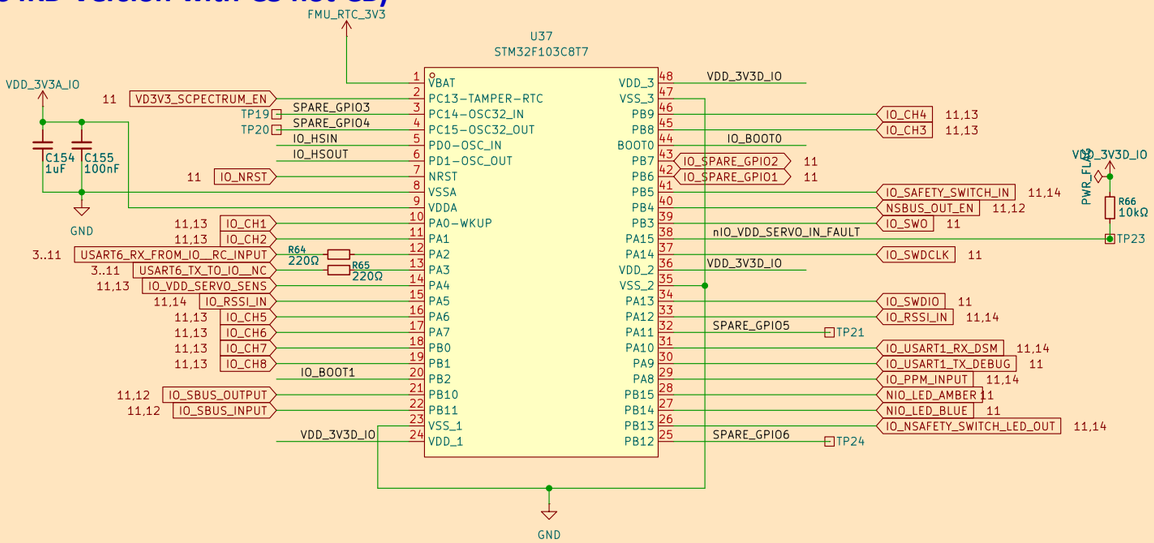

## USB POWER CONNECTORS



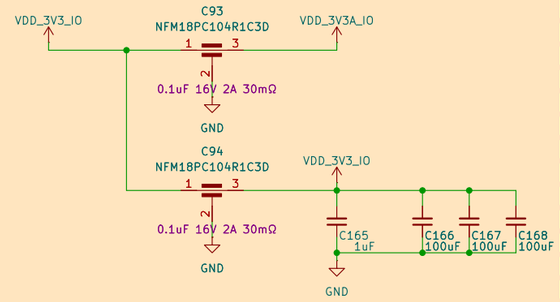




| | | | |
|---|---|---|---|
| Schematic | **Naspier Core** | Create at | 2026-09-20 |
| | | Update at | 2026-10-15 |
| Board | NASPIER_CORE_REV0_VERO | Page | **CARRIER_POWER** |
| Drawn | Ahmet Vapurcu | **Naspier Core** | |
| Reviewed | | | |
| | | Version | Size / Page 10 Total 17 |
| | | REV0 VERO | A3 / VAP-SAN |

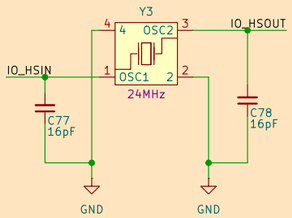
## IO F103 CHIP
## (64KB Version with C8 not CB)



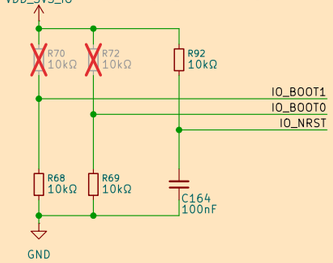

## IO POWER FILTER

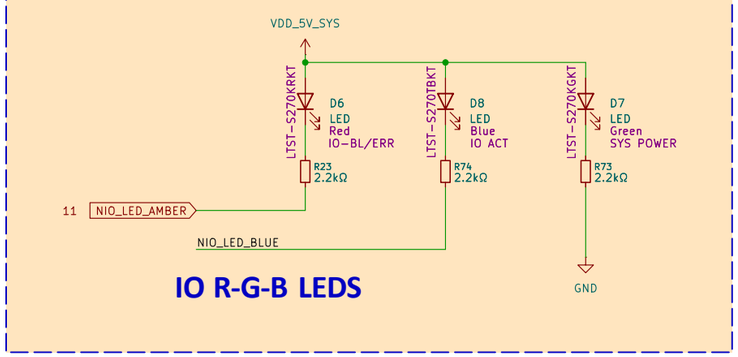



## IO CRYSTAL



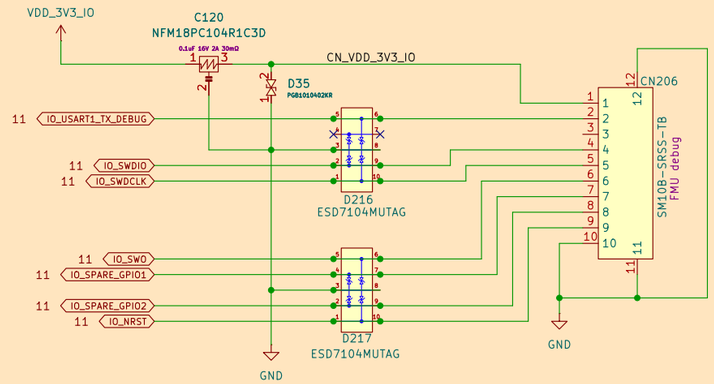

## IO BOOT-NRST



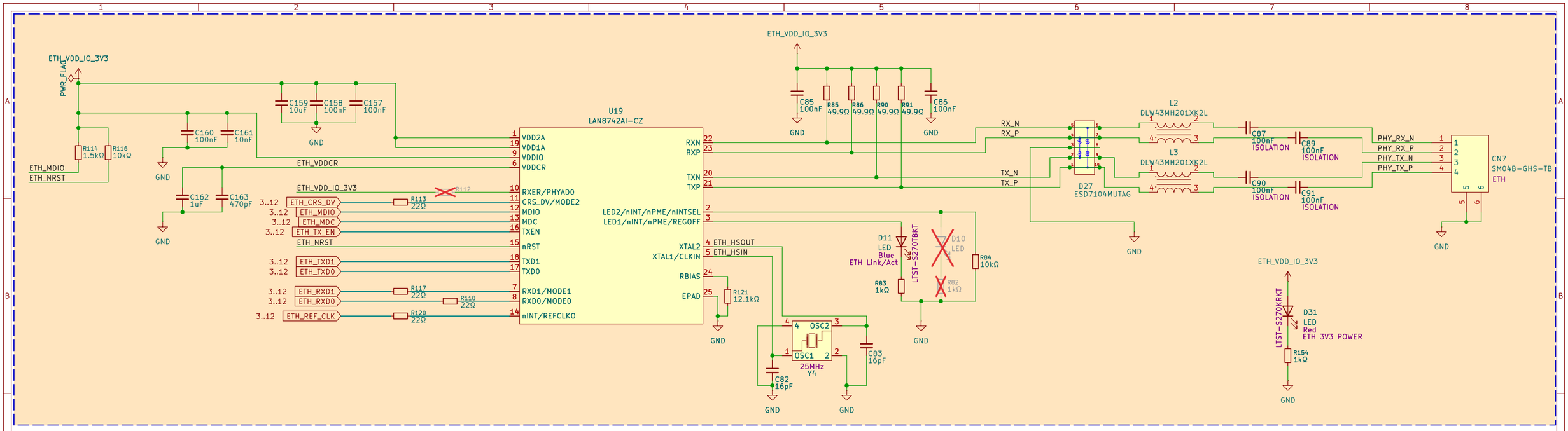




## IO R-G-B LEDS

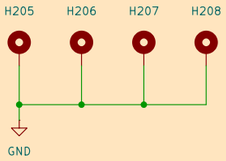
## IO DEBUG



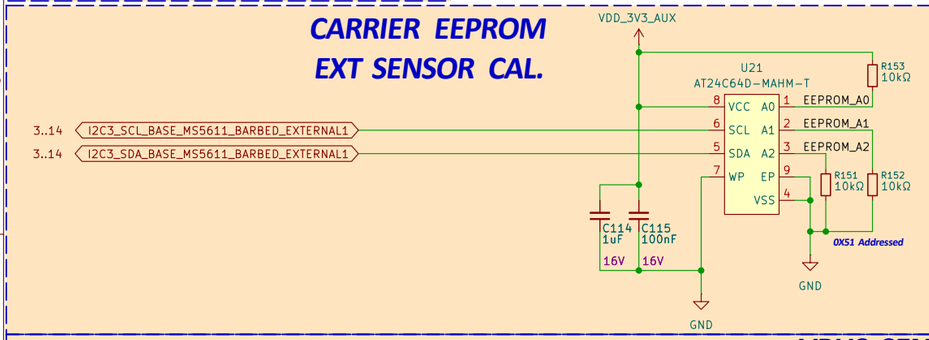

| | | | |
|---|---|---|---|
| Schematic | **Naspier Core** | Create at | 2026-09-20 |
| | | Update at | 2026-10-15 |
| Board | NASPIER_CORE_REV0_VERO | Page | **CARRIER_IO_MCU** |
| Drawn | Ahmet Vapurcu | **Naspier Core** | |
| Reviewed | | | |
| | | Version | Size | Page 11 Total 17 |
| | | REV0 VERO | A3 | VAP-SAN |



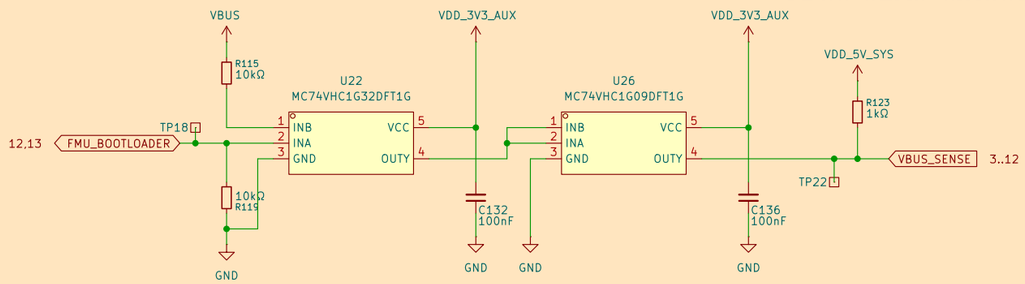

## MountingHole M3 x4 BASEBOARD HOLES



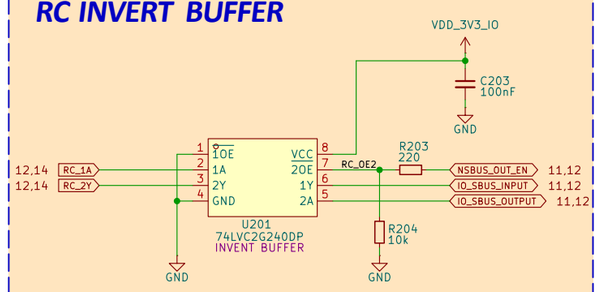

## CARRIER EEPROM EXT SENSOR CAL.



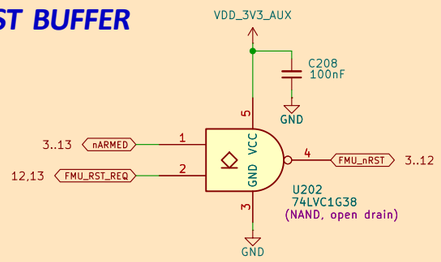

## VBUS SENS CIRCUIT




## RC INVERT BUFFER

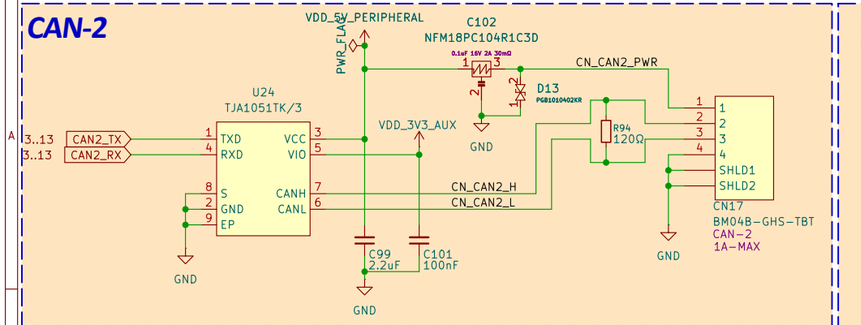



## NRST BUFFER

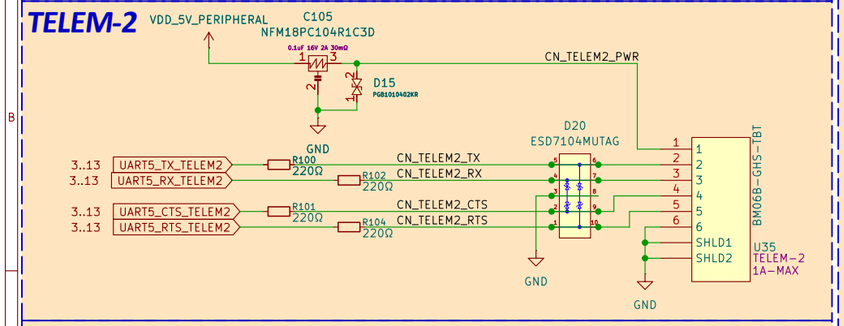



| | | | |
|---|---|---|---|
| Schematic | **Naspier Core** | Create at | 2026-09-20 |
| | | Update at | 2026-10-15 |
| Board | NASPIER_CORE_REV0_VERO | Page | **CARRIER_ELEMENTS** |
| Drawn | Ahmet Vapurcu | **Naspier Core** | |
| Reviewed | | | |
| | | Version | Size | Page 12 Total 17 |
| | | REV0 VERO | A3 | VAP-SAN |

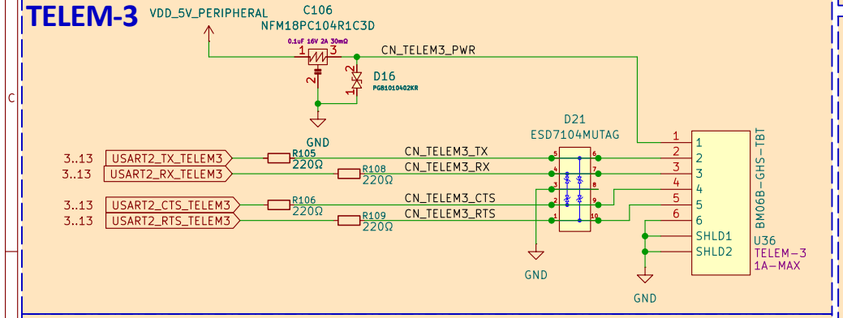
## CAN-2



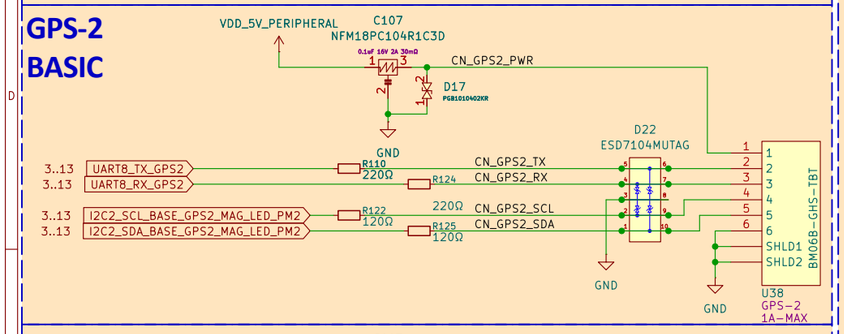

## TELEM-2



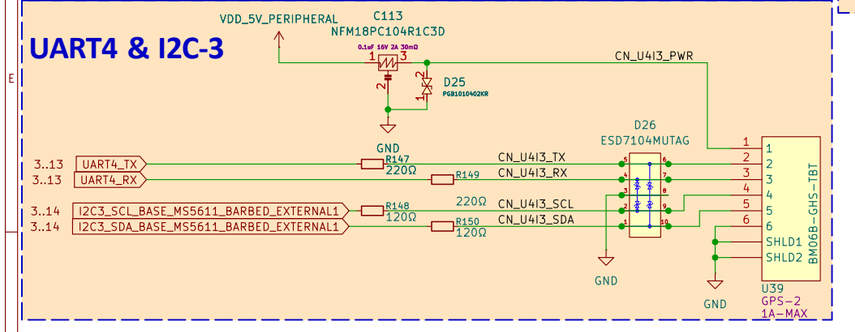

## TELEM-3



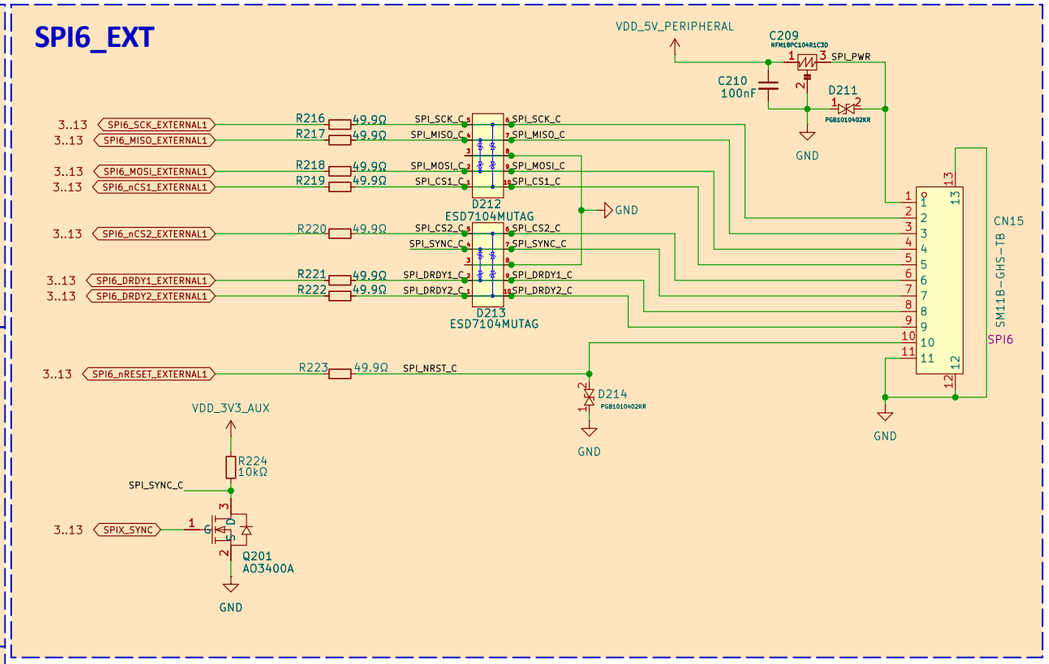

## GPS-2 BASIC



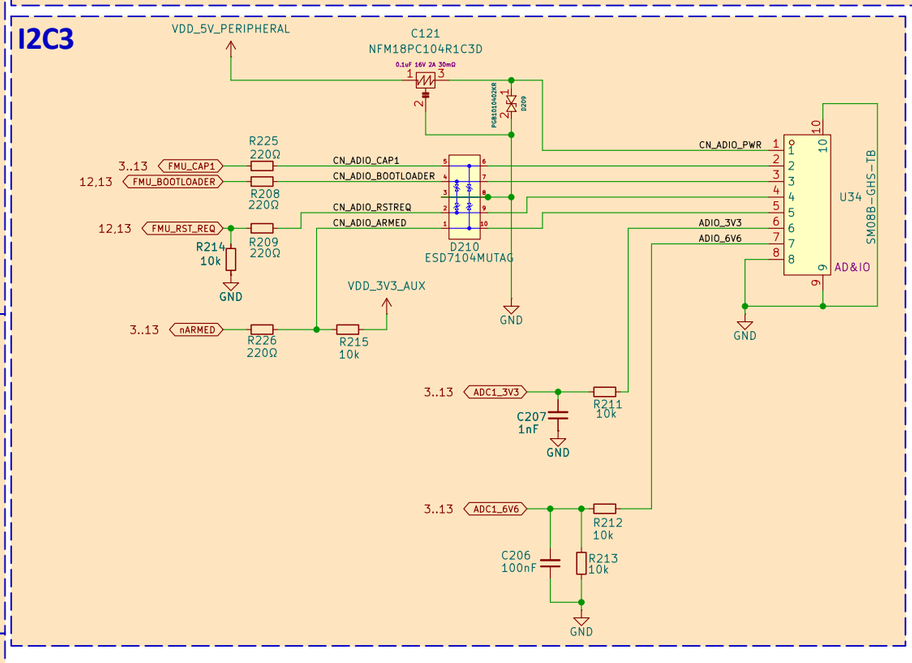

## UART4 & I2C-3



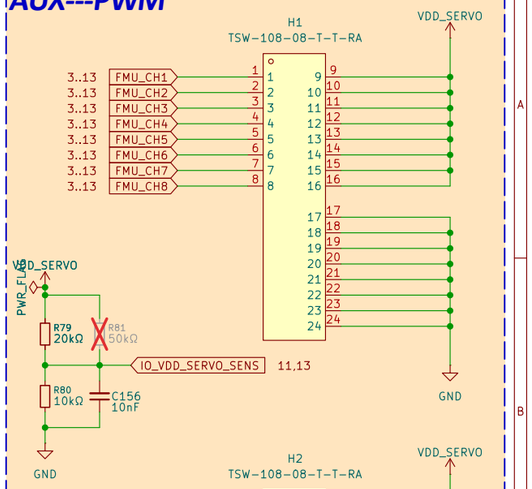

## SPI6_EXT



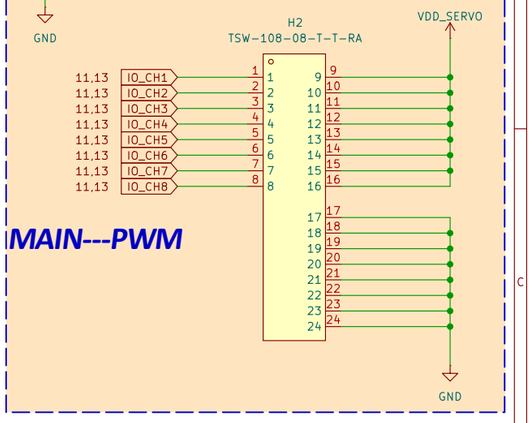

## I2C3




## AUX---PWM



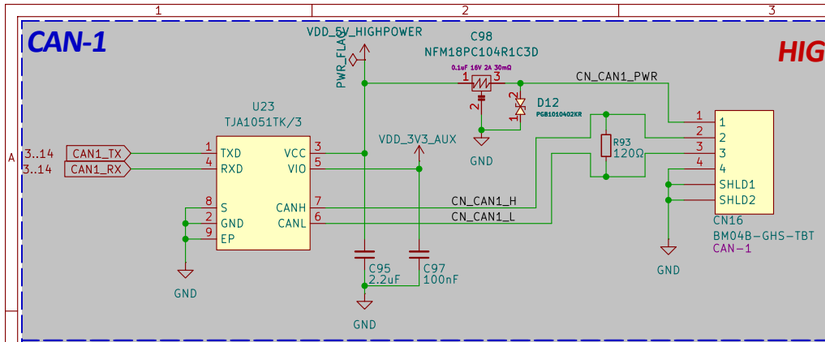

## MAIN---PWM




| | | | |
|---|---|---|---|
| Schematic | **Naspier Core** | Create at | 2026-09-20 |
| | | Update at | 2026-10-15 |
| Board | NASPIER_CORE_REV0_VER0 | Page | **CARRIER_CONN_1** |
| Drawn | Ahmet Vapurcu | **Naspier Core** | |
| Reviewed | | | |
| | | Version: REV0 VER0 | Size: A3 |
| | | Page 13 Total 17 | VAP-SAN |

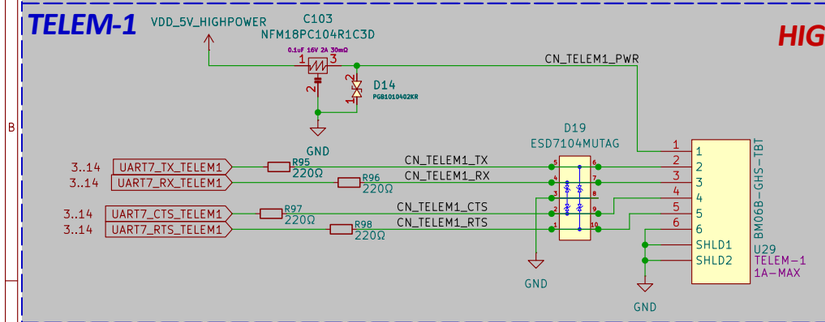
## CAN-1

HIGH POWER



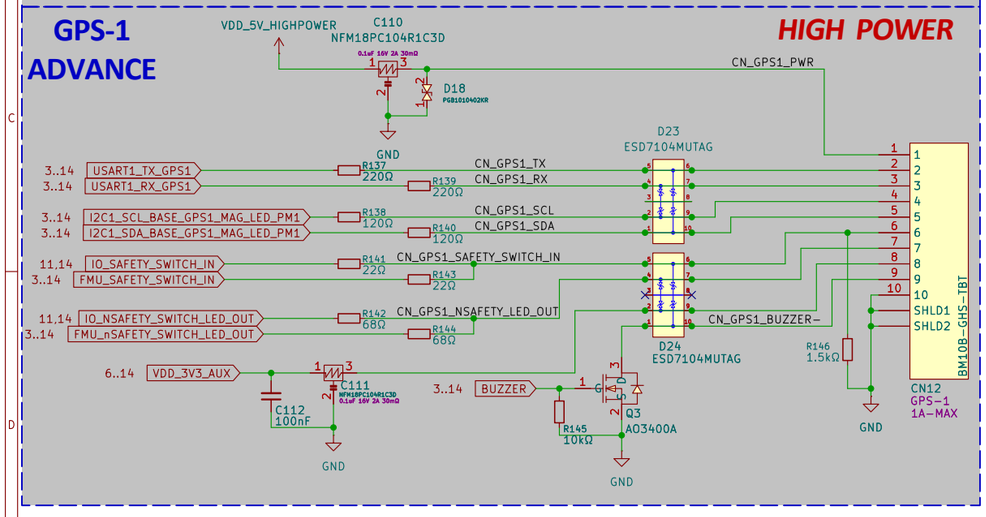

## TELEM-1

HIGH POWER

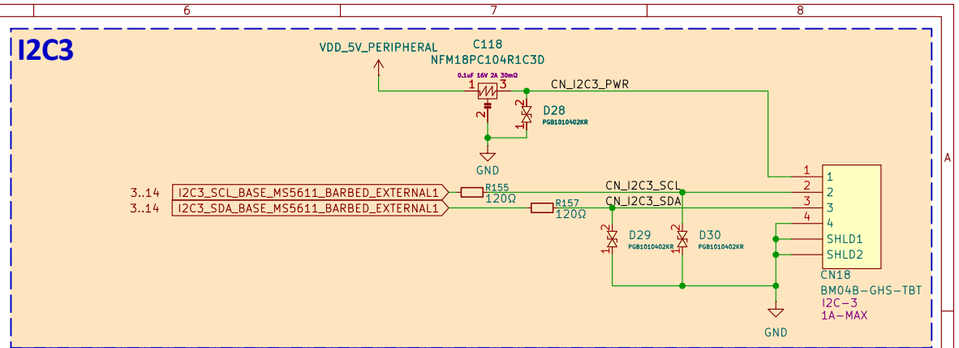



## GPS-1 ADVANCE

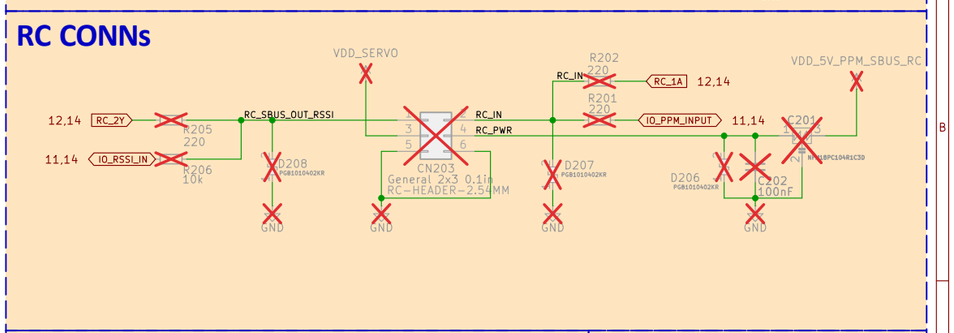
HIGH POWER



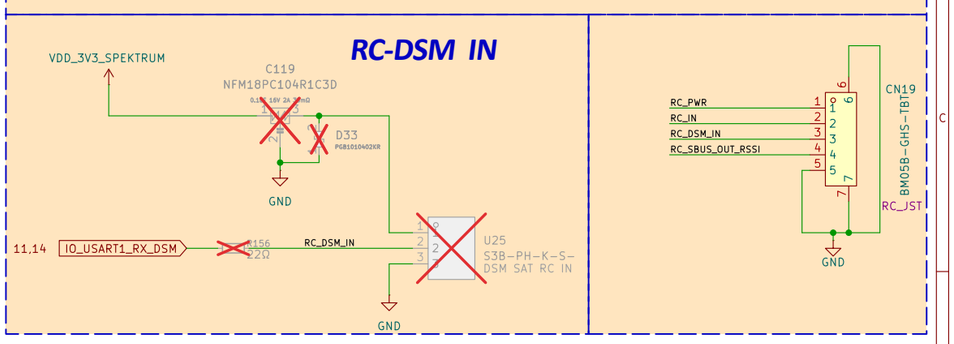

## I2C3




## RC CONNs

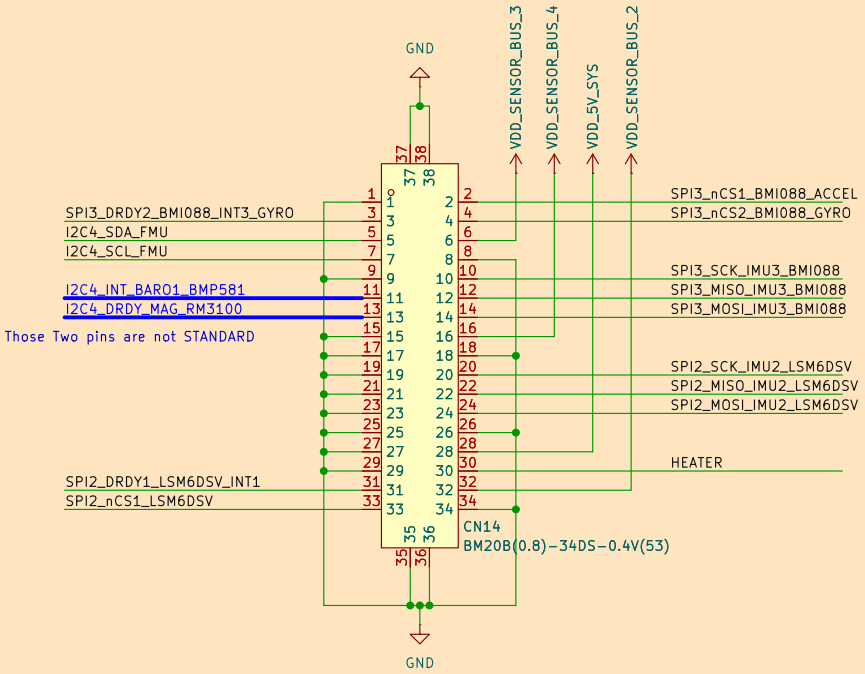


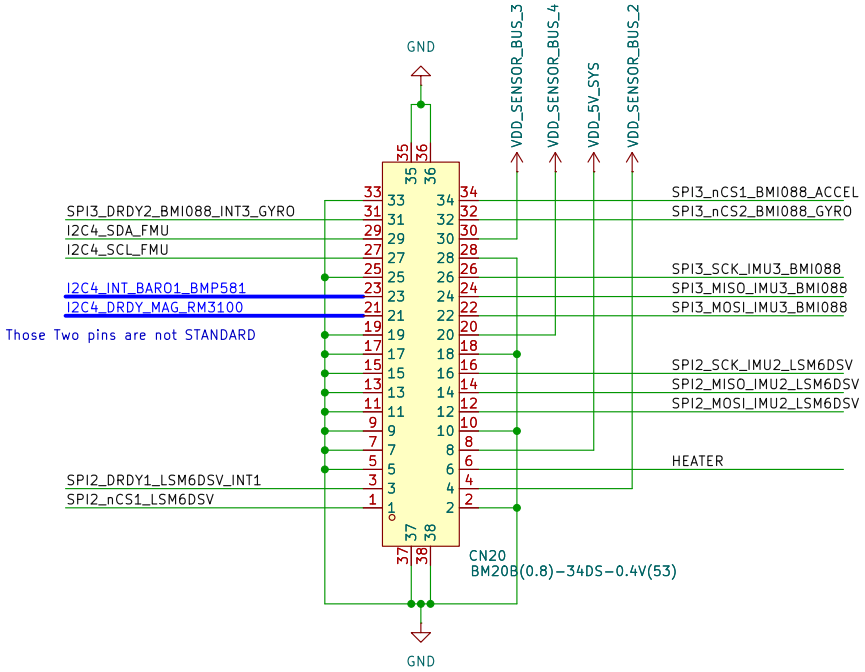

## RC-DSM IN




| | | | |
|---|---|---|---|
| Schematic | **Naspier Core** | Create at | 2026-09-20 |
| | | Update at | 2026-10-15 |
| Board | NASPIER_CORE_REV0_VER0 | Page | **CARRIER_CONN_2** |
| Drawn | Ahmet Vapurcu | **Naspier Core** | |
| Reviewed | | | |
| | | Version | REV0 VER0 |
| | | Size | A3 |
| | | Page 14 Total 17 | VAP-SAN |

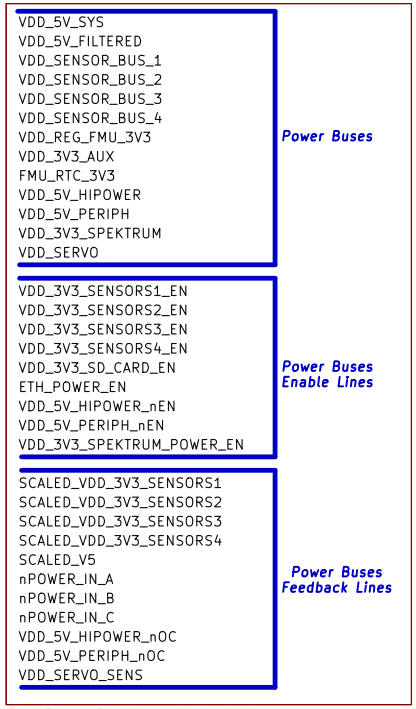
## PAB X3 34P SENSOR FFC CONNECTOR
## NON-STANDARD CONNECTOR PINOUT




| | | | |
|---|---|---|---|
| Schematic | **Naspier Core** | Create at | 2026-09-20 |
| | | Update at | 2026-10-15 |
| Board | NASPIER_CORE_REV0_VER0 | Page | |
| Drawn | Ahmet Vapurcu | **Naspier Core** | |
| Reviewed | | | |
| | | Version: REV0 VER0 | Size: A4 |
| | | Page 15 Total 17 | VAP-SAN |

POWER SECTIONS Sheet




File: Power_Sections_sch.kicad_sch

| | | | |
|---|---|---|---|
| Schematic | **Naspier Core** | Create at | 2026-09-20 |
| | | Update at | 2026-10-15 |
| Board | NASPIER_CORE_REV0_VER0 | Page | |
| Drawn | Ahmet Vapurcu | **Naspier Core** | |
| Reviewed | | | |
| | | Version | Size | Page 18 Total 17 |
| | | REV0 VER0 | A4 | VAP-SAN |

| | | | |
|---|---|---|---|
| Schematic | **Naspier Core** | Create at | 2026-09-20 |
| | | Update at | 2026-10-15 |
| Board | NASPIER_CORE_REV0_VER0 | Page | |
| Drawn | Ahmet Vapurcu | **Naspier Core** | |
| Reviewed | | | |
| | | Version | Size | Page 19 Total 17 |
| | | REV0 VER0 | A4 | VAP-SAN |